Rowan R. Terra

Ph.D. Candidate in Biological Sciences | ORISE Research Fellow
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Background

Research and development of microbiological, geochemical, and computational methods for active and abandoned mining systems. Current projects cover abandoned mine drainage remediation, critical mineral recovery, geochemical modeling, and site prioritization from field data.

Education

Duquesne University *2022–Present*
Ph.D. in Biological Sciences (anticipated October 2026) Pittsburgh, PA
Advisors: Dr. Nancy J. Trun & Dr. Djuna M. Gulliver
Committee Members: Dr. John F. Stolz & Dr. Jill A. Dembowski

Carnegie Mellon University *2023–2025*
Graduate Coursework in Biology, Chemistry, and Computer Science Pittsburgh, PA
Courses: Advanced Microbiology, Statistical Computing I and II, Transition Metal Chemistry

University of Pittsburgh *2016–2020*
M.A.T. in Science (2020) Pittsburgh, PA
Advisor: Dr. Kari Kokka
B.S. in Biological Sciences (2019)
Minors: Chemistry, Secondary Education, Religious Studies

Professional Experience

Graduate Student Researcher & Teaching Assistant *2022–Present*
Duquesne University Pittsburgh, PA

- Developed microbiological culture and enrichment methods to identify biological mining pathways from unconventional critical mineral feedstocks
- Determined microbiological mining capacity and metabolic pathways via assays and sequencing analysis
- Mentored undergraduate and graduate students
- Instructed biology lab coursework for >100 undergraduate students

Oak Ridge Institute for Science and Education (ORISE) Research Fellow *2022–Present*
U.S. Department of Energy (DOE), National Energy Technology Laboratory (NETL) Pittsburgh, PA

- Characterized critical mineral and microbiological samples using inductively coupled plasma mass spectrometry (ICP-MS), ion chromatography (IC), and scanning electron microscopy (SEM)
- Developed machine learning tools to identify critical mineral and rare earth element feedstocks using standard indicator parameters and historic survey data
- Standardized field and biological mining techniques so that data from different labs could be compared
- Created geochemical models and modeling software for the DOE Genesis Mission HERMES program
- Built DiGiTerra, a no-code machine learning tool, with DOE-NETL

Principal *2024–Present*
Terra Cognita LLC Pittsburgh, PA

- Provide statistical computing and data analysis consulting to government agencies, universities, and private companies in the US and EU
- Build machine learning models and data visualizations for client datasets
- Develop geochemical models and software for the U.S. Geological Survey (USGS) and the Office of Surface Mining Reclamation and Enforcement (OSMRE)
- Analyzed data and mentored students for a United States and German study of STEM faculty mental health with the University of Delaware and Technische Universität Braunschweig
- Certified business operating in Pennsylvania

Data Scientist and Researcher *2018–2023*
Freelance Tree Inventorying Pittsburgh, PA

- Designed custom projects, inventory maps (through ArcGIS and Google Earth Pro), and reports for clients assessing tree species, health, size, and value, among 10+ other attributes

Associate Scientist *2021–2022*
Cernostics Pittsburgh, PA

- Used immunofluorescent staining and confocal imaging on esophageal and gastric tissue samples to evaluate progression of disease and cancer
- Assisted with developing computer vision technology to reduce hand annotation on sample confocal images for prediction of future disease severity
- Conducted internal audits, developed standard operating procedures, and produced white papers

K-12 Science Educator *2019–2021*
Pittsburgh Public Schools and Oakland Catholic High School Pittsburgh, PA

- Instructed over 150 students in biological and chemical sciences coursework at two secondary schools in Pittsburgh, PA
- Adapted instruction for remote and hybrid teaching during the COVID-19 pandemic
- Developed and implemented hands-on lab approaches

Additional Research and Field Employment *2012–2019*

- Seasonal, semester-based, and part-time paid positions in pollinator field studies and microbiology, antibiotic discovery in organic chemistry laboratories, soil bacteriophage research, hydrology of South American archaeological sites, and stream assessments for trout classification and stream health

Fellowships, Grants, & Awards

Fellowships

- Oak Ridge Institute for Science and Education (ORISE) Fellowship. (2022–Present). $400,000–$500,000 total compensation. Covers Ph.D. stipend, benefits, major experimental costs, and some travel.
- American Society for Microbiology Future Leaders Mentorship Fellowship. (2025–2027).

Research Grants and Awards

- Outstanding Proposal Research Grant. Geological Society of America. (2025). $3,000.

Conference/Travel Awards

- Travel Award. Duquesne University Women in STEM. (2026). $750.
- Travel Grant. Geological Society of America. (2025). $500.
- Travel Award. Allegheny Branch of the American Society for Microbiology. (2024, 2025). ∼$250.
- Travel Grant. American Society of Reclamation Sciences. (2026). $350.

Merit Based Awards

- Outstanding Graduate Research Award. Duquesne University Graduate Research Symposium. (2024).
- Memorial Scholarship, Ph.D. Level. American Society of Reclamation Sciences. (2026).

Mentorship & Professional Associations

‡ Contributed significantly to a manuscript in preparation or under review. Mentorship covers all endeavors 2022–2026, including independent studies for United States and German STEM groups.

Name	Department	Institution	Years
J. Bowman ‡	Computer Science	Duquesne University	2026
I. Cann ‡	Biological Sciences	Duquesne University	2023–2026
C. Crawford ‡	Computer Science	Duquesne University	2026
N. Dalgaard ‡	Computer Science	Duquesne University	2026
P. DiGirolamo ‡	Computer Science	Stevens Institute of Technology	2025–2026
T. Kellman ‡	Environmental Science	Duquesne University	2026
K. Kelly ‡	Biological Sciences	Duquesne University	2025–2026
N. Lamagna ‡	Biological Sciences	Duquesne University	2022–2023
M. Lamke	Biological Sciences	Duquesne University	2025–2026
M. Meinzer	Psychologie und Psychotherapie	Technische Universität Braunschweig	2024
C. Pugh	Biological Sciences	Duquesne University	2025–2026
J. Rahman ‡	Computer Science	Duquesne University	2026
A. Strassner ‡	Biological Sciences	Duquesne University	2024–2026
R. Sumner	Biological Sciences	Duquesne University	2025
A. Thiele	Psychologie und Psychotherapie	Technische Universität Braunschweig	2024

Professional Memberships

- Duquesne University Women in STEM (WIS)
  - Small group mentor for four undergraduate students
- American Society for Microbiology (ASM)
  - Additional Allegheny Branch Membership (ABASM)
- American Chemical Society (ACS)
- American Society of Reclamation Sciences (ASRS)
- Geological Society of America (GSA)
  - Divisions: Geobiology and Geomicrobiology; Geoinformatics and Data Science; Mineralogy, Geochemistry, Petrology, and Volcanology; Economic Geology; Energy Geology
- Pittsburgh Geological Society (PGS)

Skills

Research Methods

Bench

- Aerobic and anaerobic microbiological culture, biological mining, amplicon and whole genome library preparation and sequencing, selectivity enrichment assays, liquid-liquid extraction, column chromatography

Instrumentation and Analysis

- Microscopy: SEM, SEM+EDS, Confocal, Fluorescent, Compound
- Elemental Analysis and Chromatography: IC, ICP-MS, ICP-OES, HPLC, NMR, SEM+EDS, TGA

Computational

- Languages: Python, R, C#, HTML, JavaScript
- Techniques: Machine Learning, application development, automated data-processing pipelines, HTML dashboards
- Mapping and Remote Sensing: ArcGIS, National Agriculture Imagery Program (NAIP), Google Earth Pro

Geochemical Modeling

- Software: PHREEQC
- Techniques: Aqueous speciation and saturation index modeling, thermodynamic stability (Eh-pH) diagrams with environmental and biological constraints, coupled geochemical and genomic models

Environmental

- Water/soil biological and chemical sampling, site characterization, abandoned mine drainage monitoring, mapping, satellite monitoring

Software and Data

Software

- DiGiTerra: no-code machine learning tool for exploratory analysis of environmental and geochemical data (DOE-NETL)
- terrahedral: coordination geometry comparison of metal sites in minerals and proteins

Data

- Draft genome sequences of 27 bacterial isolates from critical mineral feedstocks and abandoned mine drainage solids (NCBI GenBank)
- 16S rRNA gene amplicon dataset of Pennsylvania abandoned coal mine discharges and mine sites outside Pennsylvania
- Baas Becking (1960) Eh-pH compilation, digitized and rescued, with figure code
- Reproduction code and data for the Kentucky Hollow and computational prospecting manuscripts

Publications

Published

1. R. Terra, C. Parker, M.I. Martinez (2025). Mountain Pass Rare Earth Mine Inspires a New Generation of Economic Geoscience Professionals. *SEG Discovery*, 140.
2. R. Terra, V. Hrach (2026). Biology and Chemistry Ph.D. Student Perspectives on Graduate Teaching Assistant Experiences in Pennsylvania Identify Pedagogical Gaps. *Journal of College Science Teaching*, 55(1), 69–82.

Under Journal Review

3. R. Terra, D. Gulliver, M. Beebe, N. Trun (2026). Spatiotemporal Biogeochemistry of a Critical Mineral Rich Abandoned Mine Drainage Passive Remediation System. *Applied and Environmental Microbiology*. Manuscript AEM00153-26.
4. R. Terra, D. Gulliver, I. Cann, N. Trun (2026). Critical-Mineral Microbiomes: Draft Genome Sequences of Nine Candidate Novel Bacterial Species from California and Pennsylvania. *Microbiology Resource Announcements*. Manuscript MRA01037-26.

Under Internal Review

5. R. Terra*, P. DiGirolamo*, D. Gulliver, N. Trun (2026). DiGiTerra: A No-Code Tool for Machine Learning Exploration. *Journal of Open Source Software*. *Co-first author.
6. J. Broll, R. Terra, K. Lieb, C. Pester, S. Schaefer (2026). STEM Faculty Mental Health in Germany and the United States: The Role of National Context, Departmental Climate, and Individual Resources. *Scientific Reports*.
7. R. Terra, D. Gulliver, P. DiGirolamo, I. Cann, N. Trun (2026). Computational Prospecting for Rare Earth Element Feedstocks Across Appalachia. *Nature Geoscience*.
8. R. Terra, D. Gulliver, I. Cann, N. Trun (2026). Ubiquitous Microbiological Taxa Across Pennsylvanian Coal Mining Sites Indicate Strategies in Remediation and Metal Extraction. *mSystems*.
9. R. Terra, D. Gulliver, N. Trun (2026). Biological Mining for Critical Minerals from Secondary Feedstocks is Enhanced by Co-Contaminant Enrichments and Synthetic Communities.

10. N. Dalgaard*, J. Rahman*, R. Terra, T. Kellman, K. Kelly, D. Gulliver, N. Trun (2026). AMD Remediation Feature Identification via Computer Vision. *Reclamation Sciences*. *Co-first author.
11. R. Terra, D. Gulliver, N. Dalgaard, J. Rahman, N. Trun (2026). Computer Vision for Identification of Active, Abandoned, and Remediated Coal Mining Sites. *Computers and Geosciences*.
12. R. Terra, D. Gulliver, C. Crawford, J. Bowman, N. Trun (2026). Enhancing Experimentation through Thermodynamic Stability Diagrams with Environmental and Biological Constraints.
13. R. Terra, D. Gulliver, N. Trun (2026). terrahedral: A Tool for Coordination Comparisons Between Mineralogy and Proteins. *SoftwareX / Journal of Open Source Software*.
14. R. Terra, D. Gulliver, K. Kelly, N. Trun (2026). Comparative Metal Coordination in Microbial Proteins and Minerals Using terrahedral. *Computers and Geosciences*.
15. R. Terra, D. Gulliver, N. Trun (2026). The Microbiology of Mountain Pass Rare Earth Mine. *Frontiers in Microbiology*.
16. A. Strassner, R. Terra, N. Trun (2026). The Co-occurrence of Antibiotic, Zinc, and Cobalt Tolerances from Several Coal Mining Isolates. *Journal of the Pennsylvania Academy of Science*.
17. D. Moses, R. Terra, N. Trun, B. Porter (2027). Evaluation of Diatom Diversity Across Abandoned Mine Drainage Sites in Pennsylvania. *Applied and Environmental Microbiology*.

Contributor Roles Taxonomy (CRediT) Author Contributions by R. Terra

Contributions and expected contributions are indicated by "X".

Paper	Conceptualization	Data Curation	Formal Analysis	Funding Acquisition	Investigation	Methodology	Project Administration	Resources	Software	Supervision	Validation	Visualization	Writing: Original Draft	Writing: Review & Editing
1. Mountain Pass REE Mine	X	X			X							X	X	X
2. GTA Pedagogical Gaps	X	X	X	X	X	X	X	X	X		X	X	X	X
3. Spatiotemporal Biogeo. AMD	X	X	X		X	X			X		X	X	X	X
5. DiGiTerra: No-Code ML	X	X				X	X		X		X	X	X	X
6. STEM Faculty Mental Health		X	X		X	X			X					X
7. Comp. Prospecting REE	X	X	X		X	X			X		X	X	X	X
8. Ubiquitous Microbial Taxa	X	X	X		X	X			X		X	X	X	X
9. Biological Mining Co-Contam.	X	X	X		X	X			X		X	X	X	X
10. AMD CV Feature ID	X	X	X		X	X	X		X	X	X	X	X	X
11. CV Mining Sites	X	X	X		X	X			X		X	X	X	X
12. Thermo. Stability Diagrams	X	X	X		X	X	X		X	X	X	X	X	X
13. terrahedral: Tool	X	X	X		X	X			X		X	X	X	X
14. Metal Coord. via terrahedral	X	X	X		X	X			X		X	X	X	X
15. Mountain Pass Microbiology	X	X	X	X	X	X			X		X	X	X	X
16. Antibiotic/Zn/Co Tolerances	X	X	X		X	X	X			X	X	X		X

Professional Presentations & Invited Panels

⋆ Invited ‡ Oral

- ‡ "A Microbiological Basis for Critical Mineral and Material Acquisition from Abandoned Coal Mines in Pennsylvania." Ph.D. Dissertation Defense. Duquesne University, Department of Biological Sciences. Pittsburgh, PA. October 19, 2026 (upcoming). 60-minute oral presentation.
- ⋆‡ "The Mineral within the Protein: Predicting Structural and Functional Properties of Proteins from Mineralogical Scaffolds." Geological Society of America Connects 2026 Conference. Denver, CO. October 2026.

20-minute oral presentation.

- ‡ "Governed by Genomes and Bound by Eh-pH: Guiding Microbial Communities for Recovery of Critical Minerals." Geological Society of America Connects 2026 Conference. Denver, CO. October 2026. 20-minute oral presentation.
- ‡ "Computational Approaches Assist Prioritizing Remediation Efforts and Recovering Critical Minerals from Mining-Impacted Systems." American Society of Reclamation Sciences 2026 National Meeting. Laramie, WY. June 2026. 30-minute oral presentation.
- ‡ "Leveraging native microbial communities for reclamation, remediation, and critical mineral recovery." American Society of Reclamation Sciences 2026 National Meeting. Laramie, WY. June 2026. 30-minute oral presentation.
- ⋆‡ "ACS Graduate Community Professional Panel." Carnegie Mellon University Chapter of the American Chemical Society. Pittsburgh, PA. April 2026. 60-minute oral presentation and discussion.
- ⋆‡ "Biology and Chemistry Ph.D. Student Perspectives on Graduate Teaching Assistant Experiences in Pennsylvania Identify Pedagogical Gaps." Duquesne University, Center for Teaching Excellence. Pittsburgh, PA. March 2026. 60-minute oral presentation and article study. Digital.
- ⋆‡ "Successes and Challenges in Machine Learning for Critical Mineral Recovery in the Appalachian Coal Basin." Department of Energy, Multi-National Laboratory HERMES AI Initiative. February 2026. 20-minute oral presentation and study discussion.
- ‡ "Recursive Modeling Connects Geochemistry and Genomics in Critical Minerals Characterization." Allegheny Branch of the American Society for Microbiology Conference. California, PA. November 2025. 15-minute oral presentation.
- ⋆ "Computational Biogeochemistry Assists in Critical Mineral Recovery from Abandoned Coal Mine Drainage and Associated Precipitants." Carnegie Museum of Natural History, WILDS Symposium. October 2025. Poster
- ⋆‡ "Microbiological Mining of Critical Minerals is Altered by Community Modifying Metal Enrichments." Geological Society of America Connects 2025 Conference. San Antonio, TX. October 2025. 15-minute oral presentation.
- ‡ "Recursive Modeling Connects Geochemistry and Genomics in Critical Minerals Characterization." Geological Society of America Connects 2025 Conference. San Antonio, TX. October 2025. 15-minute oral presentation.
- ‡ "Computational Prospecting for Rare Earth Element Feedstocks in Pennsylvania." American Chemical Society Fall 2025 Conference. Washington, DC. August 2025. 15-minute oral presentation.
- ⋆‡ "Leveraging Computational Methods in Biogeochemistry: Consolidating Sampling Efforts in the Search for REE Feedstocks." Appalachian Region Technology Transfer Critical and Rare Earth Minerals. Digital. June 2025. 45-minute oral presentation.
- "Targeted Biomining and Machine Learning Approaches in Critical Minerals Revealed by a Biogeochemical Survey." Rustbelt Microbiome Conference. Pittsburgh, PA. November 2024. Poster.
- "Targeted Biomining and Machine Learning Approaches in Critical Minerals Revealed by a Biogeochemical Survey." Geological Society of America Connects 2024 Conference. September 2024. Anaheim, CA. Poster.
- ‡ "Spatiotemporal Stratification of Critical Materials and Minerals at an Acidic AMD Passive Remediation System." Allegheny Branch of the American Society for Microbiology Conference. Greensburg, PA. September 2024. 12-minute oral presentation.
- ‡ "Spatiotemporal Stratification of Critical Materials and Minerals at an Acidic AMD Passive Remediation System." Duquesne University Graduate Research Symposium. Pittsburgh, PA. March 2024. 12-minute oral presentation. *Recipient of the Outstanding Graduate Research Award.*
- "Using Mixed Microbial Communities to Solubilize Critical Materials from Abandoned Mine Drainage Remediation System." Geological Society of America Connects 2023 Conference. Pittsburgh, PA. October 2023. Poster.
- "Profiling Bacterial Communities for Future Manipulation of Abandoned Mine Drainage Contaminants." Duquesne Graduate Research Symposium. Pittsburgh, PA. March 2023. Poster.